

Chasing the “Nines” in Deep RL

Project 15

Advanced Deep Learning for Robotics, TUM

I. OBJECTIVE

The objective of this project is to explain and reduce the gap between high average return and near-perfect task success in deep reinforcement learning. ADLR Project 15 asks how to move from 90% success to 99%, 99.9%, and beyond on continuous-control tasks where optimal controllers are known to exist, such as Pendulum and CartPole-Swingup [1]. This matters for robotics because average return hides rare failures: a controller that usually works but sometimes collapses is not reliable enough for deployment. Our hypothesis is that the last failures are not caused only by insufficient exploration or too little data. They also arise from optimization pathologies in neural RL, especially critic instability, uncontrolled feature and weight norm growth, plasticity loss, and unstable effective learning rates. We will test whether a SAC agent can close this last-nine reliability gap (using mainly simbaV2-style changes, and other methods if needed and time permits).

II. RELATED WORK

Soft Actor-Critic is a suitable base algorithm because it supports continuous actions and combines entropy-regularized exploration with off-policy replay [2]. CleanRL is a practical implementation base because its single-file SAC code exposes the actor, critic, replay buffer, and update logic clearly enough to modify and instrument [3]. However, standard SAC evaluations usually emphasize mean return rather than strict reliability across seeds.

The key prior work is SimbaV2 [4]. It argues that RL scaling often fails because non-stationary data induces overfitting and unstable optimization. SimbaV2 addresses this with hyperspherical feature normalization, unit-norm weight projection with learnable scaling, and a distributional critic with reward scaling. Using SAC as the base algorithm, it reports state-of-the-art results on 57 continuous-control tasks and improved scaling with critic width, critic depth, and update-to-data ratio. This directly matches Project 15 because SimbaV2 explicitly stabilizes feature norms, parameter norms, gradient norms, and effective learning rate. Its limitation for our objective is that it studies broad benchmark performance, not the final reliability gap on small tasks that should be perfectly solvable. ReDo [5], Sample Weight Decay [6], and Fisher-Guided Selective Forgetting [7] provide fallback methods if SimbaV2 alone does not remove the failures.

III. TECHNICAL OUTLINE

We will start from CleanRL’s continuous-action SAC and reproduce baselines on Pendulum and CartPole-Swingup. SAC is chosen over DQN-style algorithms because the tasks have

continuous actions, and over PPO as the main algorithm because its replay buffer lets us inspect what behavior the agent has already discovered. Evaluation will report mean return, success rate, worst-seed return, collapse frequency, and the fraction of seeds reaching strict success thresholds.

The technical contribution is a controlled last-nine reliability study of SimbaV2-SAC, not only a reproduction. By interventions, we mean explicit SAC training changes that target optimization pathologies, such as normalization, weight projection, distributional critic losses, replay reweighting, or neuron recycling. The main intervention family will be SimbaV2. We will implement it incrementally: standard SAC, observation/reward normalization, hyperspherical feature normalization, unit-norm weight projection with learnable scalars, and the distributional critic. If full SimbaV2 works but a smaller subset works equally well, the project identifies the simpler recipe. If time permits, secondary methods will target remaining failures with ReDo-style neuron recycling [5], SWD replay weighting [6], FGSF [7], and diagnostic-triggered stabilization.

To disentangle exploration from optimization, we will use task-specific controls. Pendulum has dense rewards and low-dimensional dynamics, so exploration should be a smaller bottleneck. If replay already contains near-upright transitions but the final policy is unreliable, the evidence points toward optimization or plasticity. CartPole-Swingup is harder because the agent must discover swing-up behavior. For this task, we will check whether replay contains near-upright states, run easier initial-state or curriculum variants, and test fixed-replay updates after useful trajectories have been collected.

Scale will be treated as an experimental variable. We will vary interaction budget, critic width/depth, and update-to-data ratio, following SimbaV2. If more data closes the gap, the limiting factor is likely data. If larger networks or higher update ratios improve mean return while some seeds still fail, we will analyze feature rank, dormant-neuron fraction, feature/weight/gradient norms, and effective learning rate. Milestones: week 2, SAC metrics and replay inspection; week 4, baselines plus core SimbaV2 components; week 8, ablations and possibly secondary methods.

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